MILITARY STANDARD

MIL-STD-1320-285
(NAVY)

30 JANUARY 1997

TRUCKLOADING

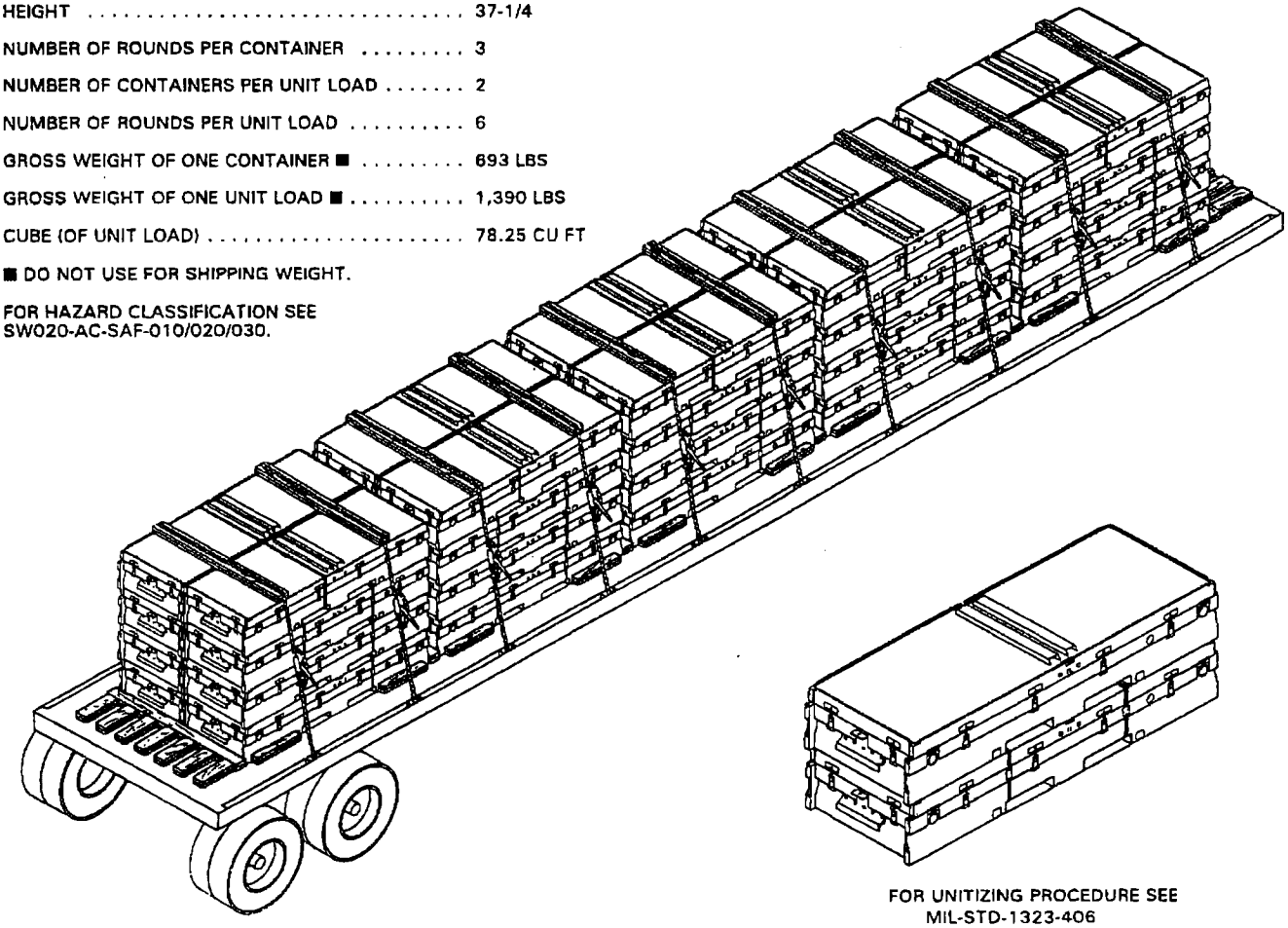
LASER GUIDED TRAINING ROUND (LGTR) IN CNU-571/E CONTAINER

UNIT LOAD DATA

LENGTH	101
WIDTH	35
HEIGHT	37-1/4
NUMBER OF ROUNDS PER CONTAINER	3
NUMBER OF CONTAINERS PER UNIT LOAD	2
NUMBER OF ROUNDS PER UNIT LOAD	6
GROSS WEIGHT OF ONE CONTAINER ■	693 LBS
GROSS WEIGHT OF ONE UNIT LOAD ■	1,390 LBS
CUBE (OF UNIT LOAD)	78.25 CU FT

■ DO NOT USE FOR SHIPPING WEIGHT.

FOR HAZARD CLASSIFICATION SEE SW020-AC-SAF-010/020/030.



FOR UNITIZING PROCEDURE SEE MIL-STD-1323-406

AMSC NO.-NA

NOTES:
1. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED TRUCKLOADING, CONTAINERLOADING, AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236.

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ORIGINATOR 5 FEB 97
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NAVAL WEAPONS STATION EARLE
COLTS NECK, NEW JERSEY

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MIL-STD-1320-285 (NAVY)

GENERAL NOTES:

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING LASER GUIDED TRAINING ROUNDS PACKED IN THE CNU-571/E SHIPPING AND STORAGE CONTAINER. CONTAINERS ARE UNITIZED INTO 2-CONTAINER UNIT LOADS IN ACCORDANCE WITH MIL-STD-1323-406 (NAVY).
2. THE PROCEDURES DESCRIBED HEREIN ARE INTENDED FOR 48 FT AND SHORTER FLATBED TRAILERS AND TRUCKS. DO NOT USE TRAILERS WITH ALL METAL FLOORS.
3. THE MAXIMUM NUMBER OF CONTAINERS:
   A) SIDE LOADING:
      THE MAXIMUM NUMBER OF CONTAINERS THAT CAN BE SIDE-LOADED ONTO A 48 FT TRAILER IS 40 CONTAINERS (FIVE STACKS WITH 4 UNIT LOADS EACH). FOR TRAILERS LESS THAN 48 FT, THE MAXIMUM NUMBER OF CONTAINERS DEPENDS UPON THE LENGTH OF THE TRAILER. THERE SHALL BE SUFFICIENT SPACE AT BOTH FORWARD AND AFT ENDS OF THE LOAD TO APPLY FLOOR BLOCKING SHOWN IN DETAIL A. TO FREE ADDITIONAL SPACE, CONTAINERS MAY BE POSITIONED AGAINST A 2 X 6 X 72 CROSSMEMBER, ITEM 3, NAILED TO TRAILER FLOOR AGAINST THE TRAILER'S FORWARD BULKHEAD, THUS ELIMINATING THE NEED FOR BACKUP CLEATS, ITEM 8, IN FRONT OF THE LOAD. THE BULKHEAD MUST BE STRONG ENOUGH TO WITHSTAND THE SUBSTANTIAL FORCE IT MAY RECEIVE DURING EMERGENCY BRAKING OR DOWNHILL TRAVEL.
   B) END LOADING:
      THE MAXIMUM NUMBER OF CONTAINERS THAT CAN BE END-LOADED ONTO A 48 FT x 102 IN. TRAILER IS 50 CONTAINERS AS SHOWN ON PAGE 7. THERE SHALL BE SUFFICIENT SPACE AT BOTH FORWARD AND AFT ENDS OF THE LOAD TO APPLY FLOOR BLOCKING SHOWN ON PAGES 7, 8 & 9.
4. UNLESS OTHERWISE SPECIFIED, DUNNAGE LUMBER, STEEL STRAPPING, STRAP SEALS, AND NAILS SHALL BE AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY).
5. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS, AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
6. AFTER THE BLOCKING, BRACING, AND TIEDOWNS HAVE BEEN INSPECTED, THE CONTAINERS SHALL BE COMPLETELY COVERED WITH FIRE RESISTANT, WATERPROOF TARPAULINS. THE TARPAULINS MAY BE UNDER THE TIEDOWNS AND THE PROTECTOR BOARDS.
7. AFTER THE TARPAULINS ARE IN PLACE, ATTACH THE PROPER EXPLOSIVES PLACARDS TO BOTH SIDES, FRONT, AND BACK OF THE TRAILER. ATTACH THE SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA ON THE BACK DECK OF THE TRAILER.
8. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY).

SIDE LOADING NOTES:

1. CHAINS AND STEEL STRAPPING MAY BE USED INTERCHANGEABLY ON A ONE-TO-ONE BASIS FOR TIEDOWNS.
2. STEEL STRAPPING USED AS TIEDOWNS SHALL BE 2 X .050 AND SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).
3. CHAINS, FITTINGS, AND LOAD BINDERS SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY), EXCEPT THAT THE CHAIN/GRAB HOOKS SHALL BE ATTACHED TO THE STAKE POCKETS (NOT AROUND THE RUB RAIL).
4. PROTECTOR BOARDS (DOUBLED 2 X 6'S) SHALL BE USED UNDER THE TIEDOWNS. WHEN USING STEEL STRAPPING, THE STRAPS SHALL BE SECURED TO THE PROTECTOR BOARD USING 2-INCH STRAPPING STAPLES. WHEN USING CHAIN, THE CHAINS SHALL BE SECURED TO THE PROTECTOR BOARD BY DRIVING A 10d NAIL THROUGH THE CHAIN LINK AND CLINCHING THE NAIL OVER THE CHAIN. FIVE STAPLES/NAILS SHALL BE USED FOR EACH TIEDOWN.
5. AFTER EACH STACK OF CONTAINERS IS LOADED ONTO THE TRAILER, THE TWO CONTAINERS IN THE UPPERMOST LAYER SHALL BE BOUND TOGETHER BY ONE CROSS STRAP, ITEM 7, AS SHOWN ON PAGE 4.
6. PRIOR TO LOADING THE TRAILER AND DURING PRELOADING INSPECTION REQUIRED BY NAVSEA OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS, AND LOAD BINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR, AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS, FITTINGS, AND LOAD BINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 13-T OF DD FORM 626, OCT. 95. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOAD BINDER.

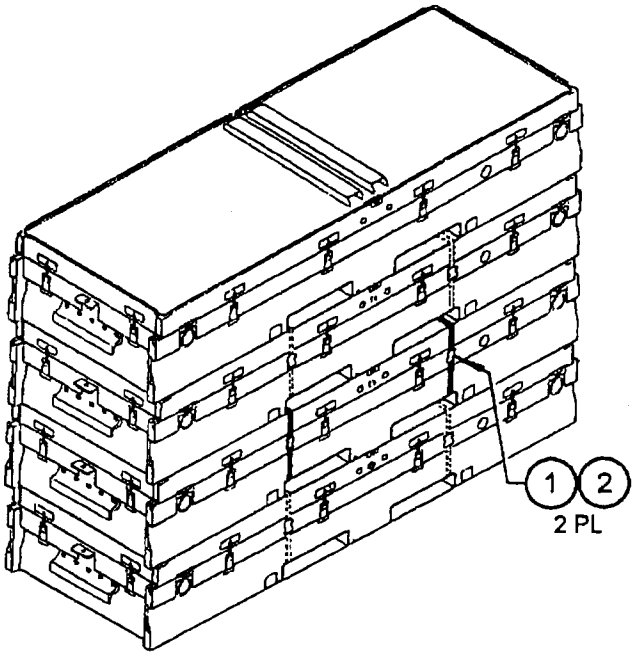
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END LOADING NOTES:

1. CONTAINER ORIENTATION:
   WHILE KEEPING THE UNIT LOAD'S FWD END TO THE LEFT SIDE OF THE TRAILER, LOAD THE FIRST STACK ON THE TRAILER SO THAT THE LOWER CONTAINER'S UNDERSIDE BETWEEN THE FORKPOCKETS IS CAPTURED BY THE INTERNAL BRACE.
   LOAD THE REMAINING UNIT LOADS ONTO THE TRAILER WHILE ALTERNATING THE ORIENTATION OF THE FWD. END OF THE UNIT LOAD BETWEEN THE LEFT AND RIGHT SIDES OF THE TRAILER FOR EACH SUCCEDING STACK.
2. STEEL STRAPPING:
   THE FIRST TWO STACKS OF FOUR CONTAINERS HIGH SHALL HAVE THE UPPERMOST LAYER BOUND TOGETHER BY TWO STEEL STRAPS, ITEM 7, AS SHOWN ON PAGE 7. REPEAT FOR THE LAST TWO STACKS.
3. WEB STRAPPING:
   ADD TWO WEB STRAPS, ITEM 10, TO ALL STACKS. SEE DETAIL D ON PAGE 7 FOR ATTACHMENT METHOD TO RUB RAIL.
   WEB STRAP ASSEMBLIES SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN JOINT DOD POLICY ON THE USE OF COMMERCIALLY AVAILABLE WEB STRAP TIEDOWNS FOR HIGHWAY SHIPMENT OF DOD MUNITIONS DATED 25 JUNE 1996, REVISION A (JOINT DOD POLICY). PRIOR TO LOADING THE TRAILER AND DURING PRELOADING INSPECTION REQUIRED BY NAVSEA OP 2165 AND REPORTED ON DD FORM 626, THE WEB STRAP ASSEMBLIES SHALL BE INSPECTED ACCORDING TO JOINT DOD POLICY. THE INSPECTOR SHALL CONFIRM THAT THE WEB STRAP ASSEMBLIES HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 13-T OF DD FORM 626, OCT. 95.
   JOINT DOD POLICY LISTS CONDITIONS WHICH REQUIRE REJECTION OF WEB STRAP ASSEMBLIES.

UNITIZING PROCEDURE FOR TWO UNIT LOADS (4 CONTAINERS):

1. CAREFULLY STACK ONE UNIT LOAD (TWO CONTAINERS) ON TOP OF ANOTHER UNIT LOAD. BE SURE TO ALIGN THE STACKING FEATURES. THE TWO UNIT LOADS SHALL HAVE BEEN UNITIZED IN ACCORDANCE WITH MIL-STD-1323-406.
2. THREAD STRAPPING, ITEM 1, THROUGH EACH FORK POCKET OF THE MIDDLE TWO CONTAINERS AS SHOWN BELOW. TENSION AND SECURE EACH STRAP WITH ONE DOUBLE NOTCHED SEAL, ITEM 2.



WARNING

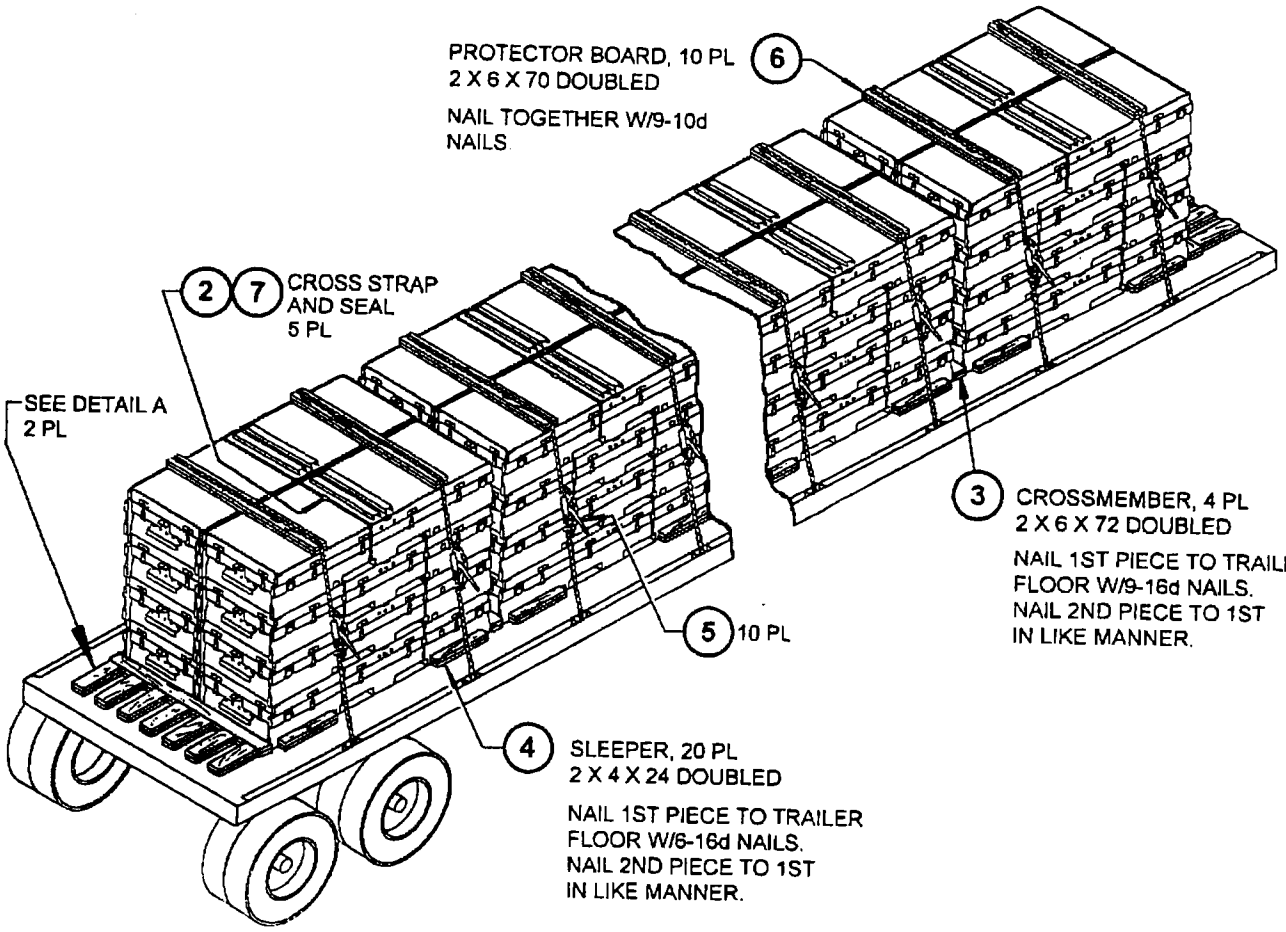
THE STRAPPING TOGETHER OF CONTAINERS TO FORM A UNIT LOAD OF CONTAINERS FOUR HIGH IS AUTHORIZED FOR THE LOADING AND UNLOADING OF TRAILERS ONLY. ANY MOVEMENT OF THESE 4-HIGH UNIT LOADS IS LIMITED TO THAT NECESSARY TO LOAD OR UNLOAD THE TRAILER.

REQ	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
AR	2	STRAP SEAL	STEEL	1-1/4" SIZE
AR	1	UNITIZING STRAP	STEEL	1-1/4 X .035 X 11 FT

LIST OF MATERIALS

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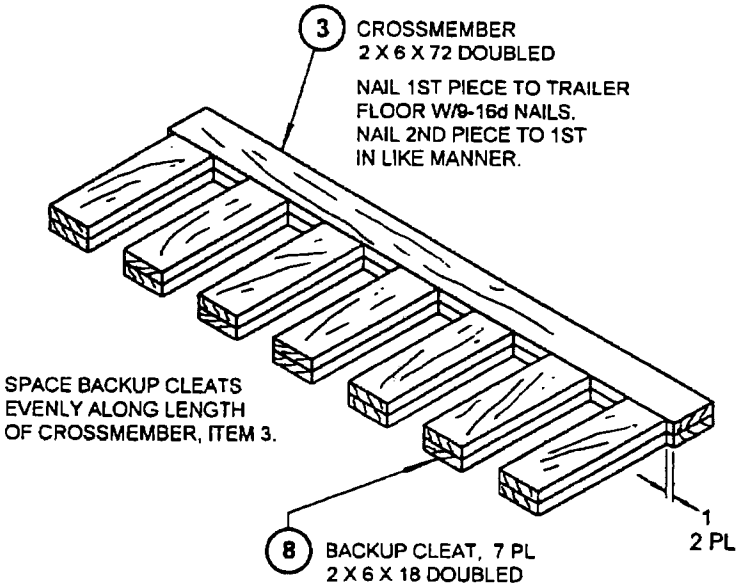
FTL 48 FT TRAILER (FLATBED)



LIST OF MATERIALS

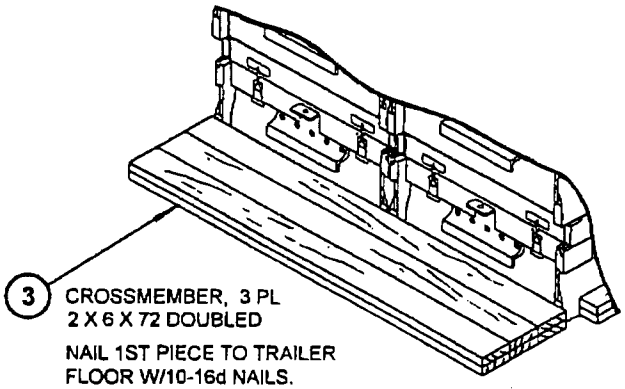
REQ	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
AR	7	CROSS STRAP	STEEL	1-1/4 X .035 X 20 FT
AR	6	PROTECTOR BOARD	WOOD	2 X 6 X LENGTH REQ'D.
AR	5	CHAIN & LOADBINDER	STEEL	
AR	4	SLEEPER	WOOD	2 X 4 X LENGTH REQ'D.
AR	3	CROSSMEMBER	WOOD	2 X 6 X LENGTH REQ'D.

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DETAIL A

FWD/REAR END BLOCKING
(SEE DETAIL B FOR ALTERNATE REAR BLOCKING)



DETAIL B

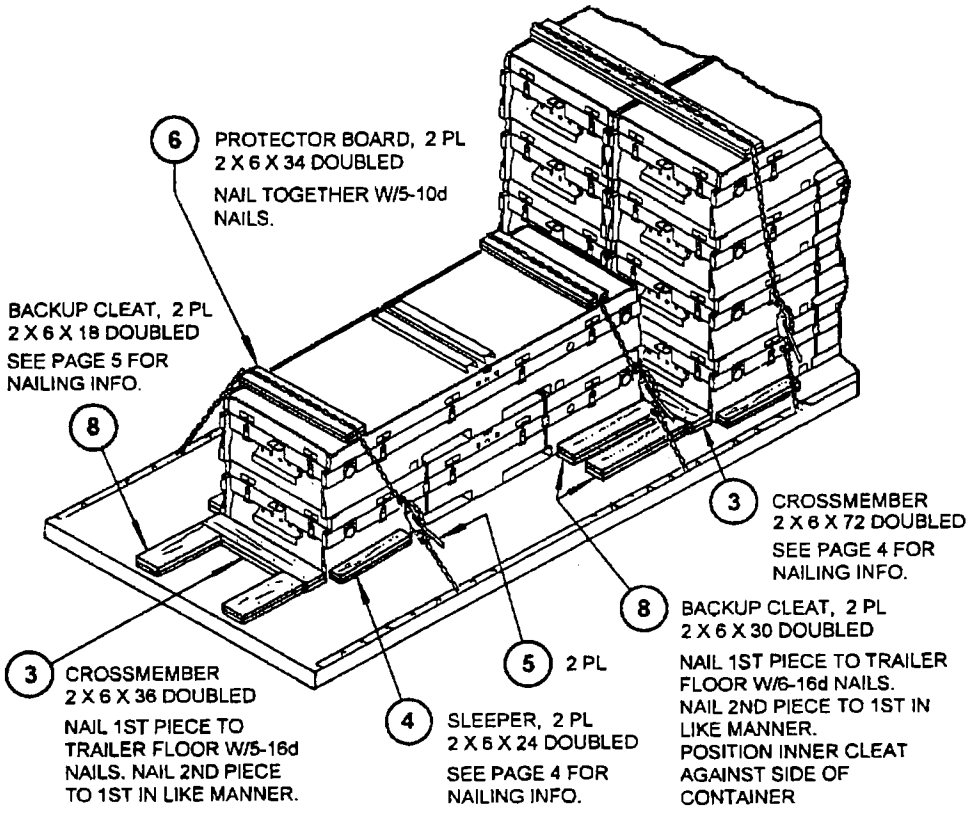
ALTERNATE REAR END BLOCKING ASSEMBLY TO BE USED WHEN THERE IS INSUFFICIENT SPACE TO APPLY DETAIL A BLOCKING. THIS METHOD SHALL NOT BE USED FOR FORWARD END BLOCKING.

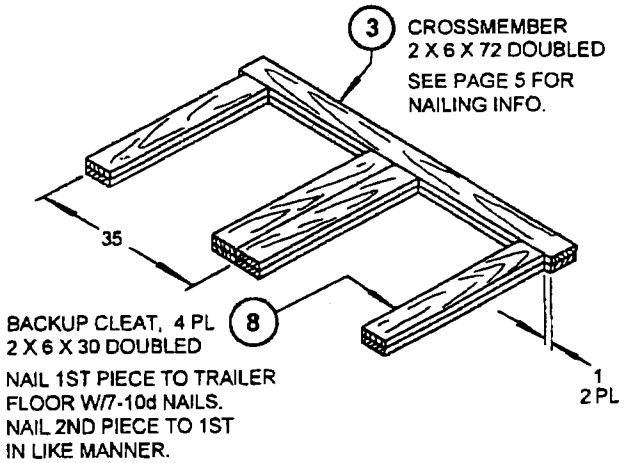
AR	B	CLEAT	WOOD	2 x 6 x LENGTH REQ'D.
REQ	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
LIST OF MATERIALS				

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LESS THAN FULL LOAD (LTL) NOTES:

1. LESS THAN FULL LOADS MAY BE ARRANGED ON THE TRAILER IN ANY COMBINATION OF STACKS CONSISTING OF ONE, TWO, FOUR, SIX, OR EIGHT CONTAINERS. STACKS OF THREE, FIVE, OR SEVEN CONTAINERS ARE NOT AUTHORIZED AS THEY WOULD BE UNEVEN STACKS. SEE DEFINITIONS CHAPTER IN MIL-STD-1320 FOR MEANING OF THE WORD "STACK".
2. FLOOR BLOCKING FOR THE INTERFACE BETWEEN A 2-WIDE STACK OF CONTAINERS AND A 1-WIDE STACK SHALL BE APPLIED AS SHOWN BELOW.
3. END BLOCKING AND TIEDOWNS FOR 1-WIDE STACKS SHALL BE APPLIED AS SHOWN BELOW.
4. THE END BLOCKING ASSEMBLY SHOWN IN DETAIL C MAY BE USED IN LIEU OF DETAIL A WHEN THERE IS SUFFICIENT SPACE. THIS ASSEMBLY MAY BE USED AT BOTH FORWARD AND AFT ENDS OF THE LOAD.





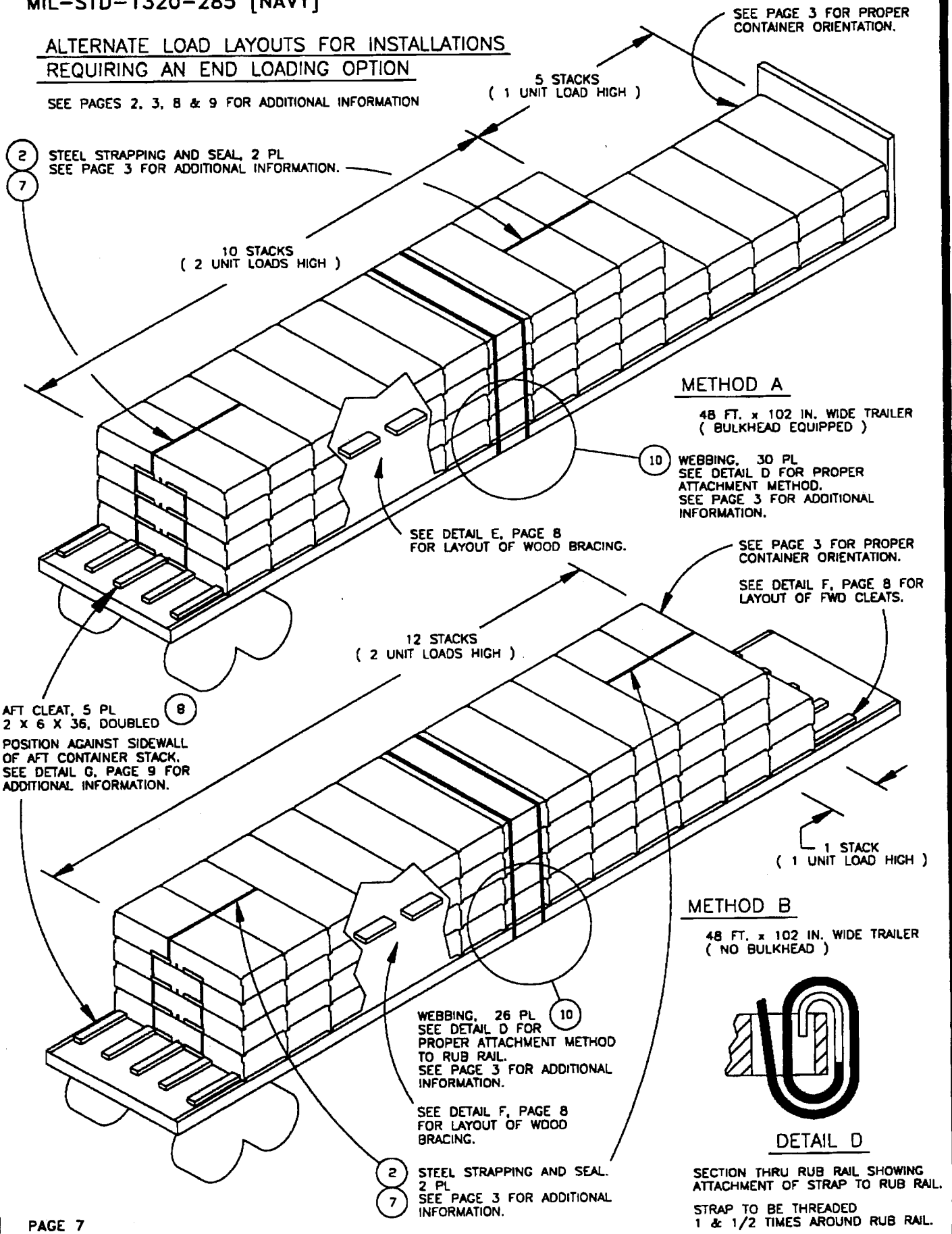
DETAIL C

LTL ALTERNATE FWD/REAR END BLOCKING ASSEMBLY

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ALTERNATE LOAD LAYOUTS FOR INSTALLATIONS REQUIRING AN END LOADING OPTION

SEE PAGES 2, 3, 8 & 9 FOR ADDITIONAL INFORMATION

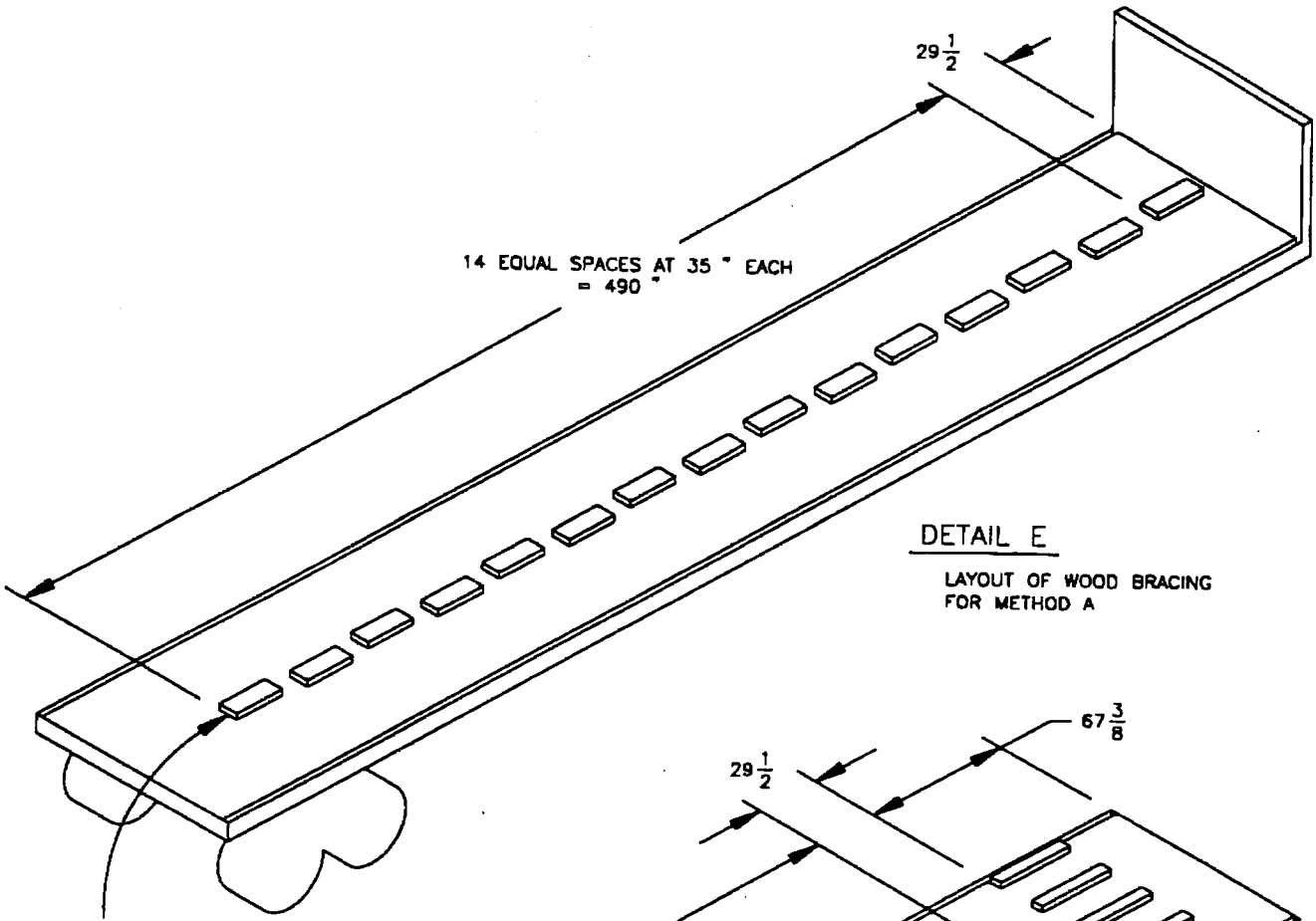


DETAIL D

SECTION THRU RUB RAIL SHOWING ATTACHMENT OF STRAP TO RUB RAIL.

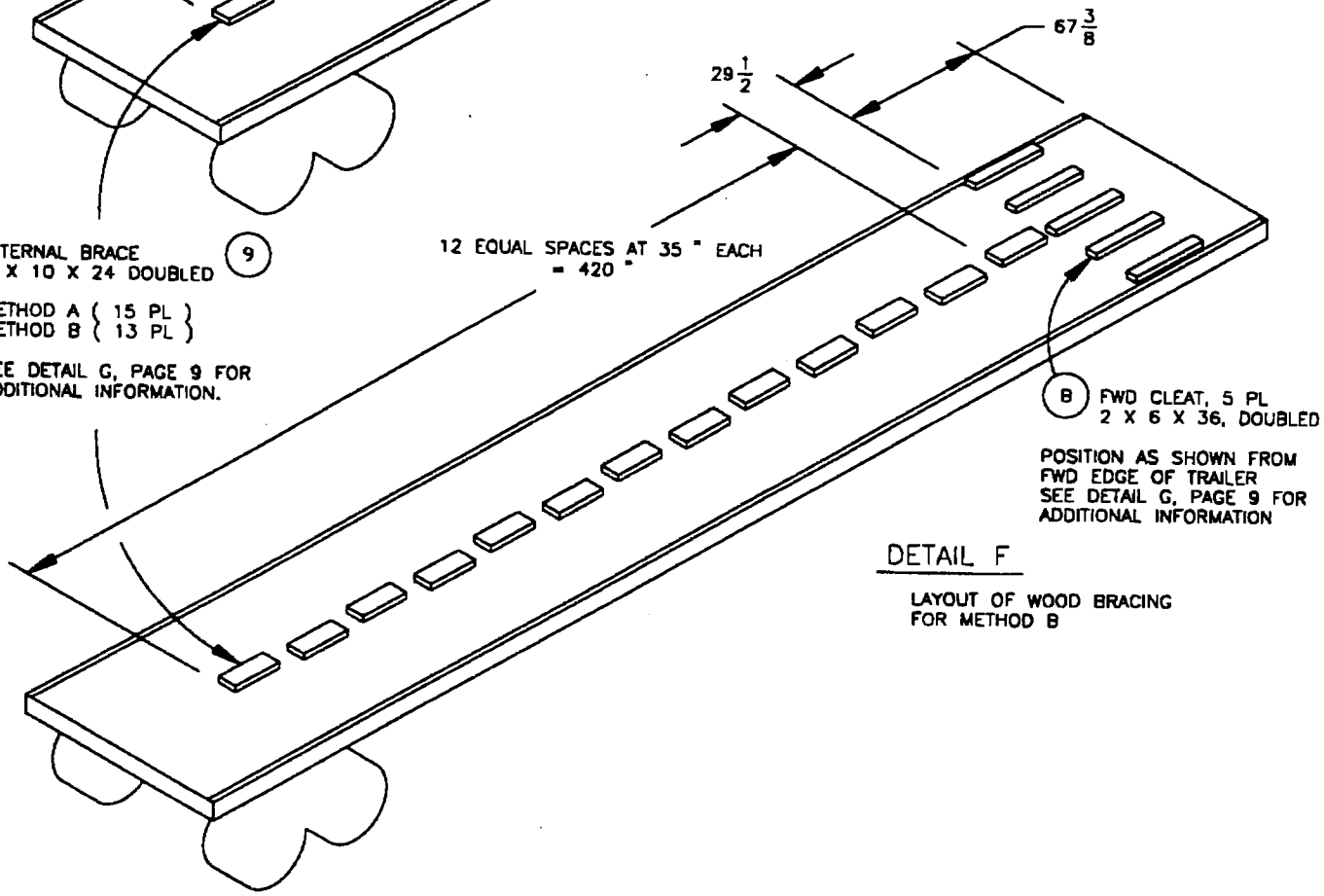
STRAP TO BE THREADED 1 & 1/2 TIMES AROUND RUB RAIL.

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DETAIL E

LAYOUT OF WOOD BRACING
FOR METHOD A



INTERNAL BRACE
2 X 10 X 24 DOUBLED (9)

METHOD A (15 PL)
METHOD B (13 PL)

SEE DETAIL G, PAGE 9 FOR
ADDITIONAL INFORMATION.

(8) FWD CLEAT, 5 PL
2 X 6 X 36, DOUBLED

POSITION AS SHOWN FROM
FWD EDGE OF TRAILER
SEE DETAIL G, PAGE 9 FOR
ADDITIONAL INFORMATION

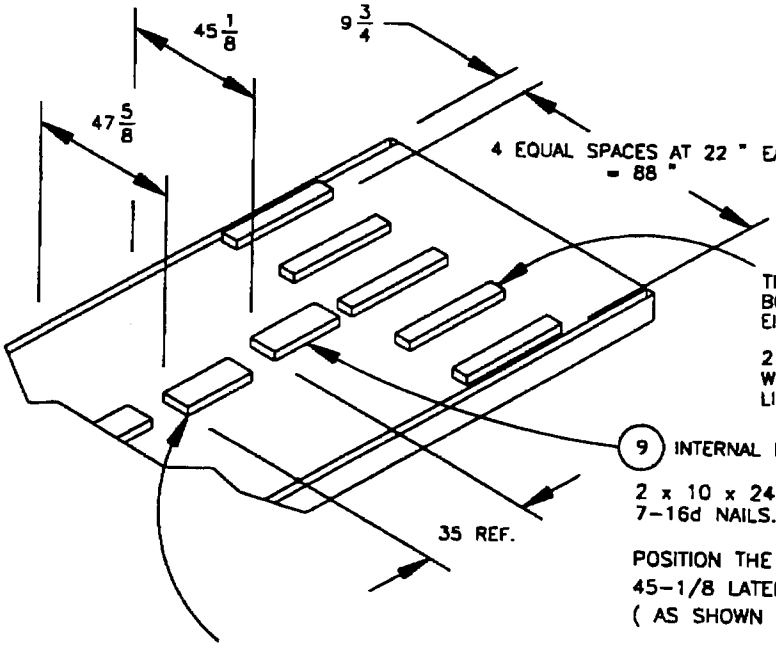
DETAIL F

LAYOUT OF WOOD BRACING
FOR METHOD B

REQ	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
AR	10	WEBBING	MIN. 1,750 LB. WORKING LOAD LIMIT	
AR	9	INTERNAL BRACE	WOOD	2 X 10 X 24

LIST OF MATERIALS

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THE 9-3/4 & 22 DIMS. SHOULD BE USED TO POSITION BOTH FWD AND AFT CLEATS WHERE APPLICABLE FOR EITHER METHOD A OR B.

2 x 6 x 36 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR WITH 11-16d NAILS. NAIL SECOND 2ND PIECE TO 1ST IN LIKE MANNER. (5 PL)

(9) INTERNAL BRACE

2 x 10 x 24 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR WITH 7-16d NAILS. NAIL SECOND 2ND PIECE TO 1ST IN LIKE MANNER.

POSITION THE FIRST INTERNAL BRACE USING THE 45-1/8 LATERAL DIM. AND THE 29 1/2 LONGITUDINAL DIM. (AS SHOWN IN EITHER DETAIL E OR F, PAGE 8).

STARTING WITH THE 47-5/8 LATERAL DIM., ALTERNATE THE LATERAL POSITIONING OF THE REMAINING INTERNAL BRACES BETWEEN 47-5/8 AND 45-1/8 UNTIL THE REQ'D. NUMBER OF BRACES IS OBTAINED. MAINTAIN 35 IN. LONGITUDINAL SPACING BETWEEN CORRESPONDING EDGES OF ALL INTERNAL BRACES.

DETAIL G

TYP. LAYOUT OF ALL CLEATS & INTERNAL BRACING

FWD. END OF METHOD B TRAILER IS SHOWN. LAYOUT OF METHOD A TRAILER IS SIMILAR.

REVIEW ACTIVITY
NAVY - OS

PREPARING ACTIVITY
NAVY - OS
PROJECT NO. 8140-0152

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